

## NIAMEY, Niger

WMO: 610520

Lat: 13.482N

Lon: 2.184E

Elev: 732

StdP: 14.31

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 60.7       | 62.6 | 8.9                           | 8.9 | 90.5 | 12.3 | 10.6 | 88.5 | 23.0                  | 76.3 | 21.1 | 77.1 | 5.6                      | 0    | 0.463 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 23.1                   | 109.0           | 68.6 | 107.5 | 68.4 | 105.8 | 68.5 | 80.6                | 91.3 | 79.7 | 90.5 | 78.9 | 90.0 | 7.5                  | 40   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 78.6                            | 152.5 | 84.5 | 77.2 | 145.5 | 84.0 | 76.6 | 142.6 | 83.7 | 44.9          | 92.0 | 43.9 | 90.8 | 43.0 | 89.8 | 86.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 20.2          | 17.5 | 15.5 | 56.4                       | 111.8 | 2.7                | 1.1 | 54.4                                               | 112.6 | 52.9       | 113.2 | 51.4       | 113.8 | 49.4       | 114.6 |    |    |
| (5)               |      |      | 42.4                       | 82.4  | 1.1                | 2.1 | 41.7                                               | 83.9  | 41.0       | 85.1  | 40.4       | 86.3  | 39.6       | 87.8  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 86.2   | 76.9  | 82.4  | 89.5  | 94.9  | 94.6  | 90.3  | 85.5  | 82.9  | 85.6  | 88.9  | 84.4  | 78.5  |
|                                                                                  | DBStd       | 6.62   | 3.96  | 4.99  | 4.13  | 3.16  | 3.61  | 4.14  | 3.80  | 3.22  | 3.67  | 2.83  | 3.38  | 3.61  |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDD50       | 13214  | 833   | 908   | 1224  | 1346  | 1382  | 1209  | 1100  | 1021  | 1067  | 1207  | 1033  | 884   |
|                                                                                  | CDD65       | 7739   | 368   | 488   | 759   | 896   | 917   | 759   | 635   | 556   | 617   | 742   | 583   | 419   |
|                                                                                  | CDH74       | 110038 | 4066  | 6360  | 11694 | 14999 | 15166 | 11539 | 8297  | 6386  | 7990  | 10823 | 7931  | 4787  |
| CDH80                                                                            | 67724       | 2034   | 3742  | 7723  | 10734 | 10736 | 7384  | 4355  | 2807  | 4177  | 6640  | 4845  | 2547  |       |
| Wind                                                                             | WSAvg       | 7.3    | 8.1   | 7.9   | 7.3   | 7.0   | 8.5   | 9.0   | 8.2   | 6.8   | 6.3   | 5.7   | 5.6   | 7.3   |
| Precipitation                                                                    | PrecAvg     | 21.2   | 0.0   | 0.0   | 0.1   | 0.3   | 1.2   | 2.9   | 5.8   | 6.9   | 3.5   | 0.6   | 0.0   | 0.0   |
|                                                                                  | PrecMax     | 34.4   | 0.0   | 0.0   | 1.7   | 2.2   | 4.3   | 8.1   | 13.0  | 12.9  | 9.4   | 3.9   | 0.2   | 0.2   |
|                                                                                  | PrecMin     | 11.5   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 1.6   | 0.6   | 0.9   | 0.0   | 0.0   | 0.0   |
|                                                                                  | PrecStd     | 5.5    | 0.0   | 0.0   | 0.3   | 0.6   | 1.1   | 1.6   | 2.2   | 2.8   | 1.9   | 0.9   | 0.0   | 0.0   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 98.9  | 105.6 | 108.2 | 111.0 | 110.8 | 107.4 | 101.8 | 97.1  | 101.1 | 104.2 | 102.6 | 98.9  |
|                                                                                  |             | MCWB   | 61.1  | 63.0  | 64.3  | 68.0  | 69.7  | 72.7  | 74.3  | 76.3  | 75.1  | 69.6  | 65.2  | 63.7  |
|                                                                                  | 2%          | DB     | 95.4  | 102.3 | 106.7 | 109.3 | 108.5 | 104.3 | 98.7  | 94.8  | 98.8  | 102.4 | 101.0 | 96.7  |
|                                                                                  |             | MCWB   | 59.8  | 62.6  | 63.9  | 67.9  | 71.1  | 73.7  | 75.4  | 76.8  | 75.9  | 69.9  | 64.8  | 62.2  |
|                                                                                  | 5%          | DB     | 93.3  | 100.1 | 105.4 | 107.7 | 107.1 | 102.4 | 96.5  | 92.9  | 96.8  | 100.8 | 100.0 | 94.8  |
|                                                                                  |             | MCWB   | 59.1  | 61.9  | 63.6  | 67.5  | 71.8  | 74.3  | 75.5  | 76.8  | 76.4  | 70.1  | 64.4  | 61.0  |
|                                                                                  | 10%         | DB     | 90.7  | 96.9  | 103.2 | 106.1 | 104.9 | 100.3 | 94.2  | 90.8  | 94.6  | 99.1  | 98.2  | 92.4  |
|                                                                                  |             | MCWB   | 58.3  | 60.6  | 63.0  | 67.6  | 72.7  | 74.6  | 75.6  | 76.6  | 76.6  | 71.2  | 63.8  | 59.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 65.0  | 69.9  | 74.4  | 79.3  | 81.5  | 80.7  | 80.9  | 81.2  | 81.9  | 80.5  | 73.9  | 66.8  |
|                                                                                  |             | MCDB   | 92.0  | 90.2  | 90.5  | 98.1  | 97.5  | 94.2  | 90.9  | 89.0  | 91.2  | 91.3  | 92.8  | 95.5  |
|                                                                                  | 2%          | WB     | 62.1  | 66.1  | 71.3  | 77.4  | 79.5  | 79.2  | 79.5  | 79.8  | 80.4  | 78.9  | 71.1  | 64.3  |
|                                                                                  |             | MCDB   | 91.5  | 94.7  | 93.5  | 96.7  | 96.2  | 92.9  | 89.2  | 87.8  | 89.7  | 90.3  | 91.7  | 92.1  |
|                                                                                  | 5%          | WB     | 60.3  | 63.3  | 67.8  | 75.9  | 78.2  | 78.3  | 78.6  | 78.9  | 79.3  | 77.8  | 68.5  | 62.3  |
|                                                                                  |             | MCDB   | 90.1  | 96.2  | 97.6  | 95.9  | 95.9  | 92.3  | 88.1  | 86.8  | 89.0  | 89.9  | 92.7  | 90.8  |
|                                                                                  | 10%         | WB     | 58.6  | 61.4  | 64.8  | 74.5  | 77.0  | 77.4  | 77.7  | 78.1  | 78.5  | 76.7  | 66.0  | 60.5  |
|                                                                                  |             | MCDB   | 87.9  | 95.1  | 99.5  | 95.1  | 95.2  | 91.7  | 87.5  | 85.8  | 88.3  | 89.5  | 91.2  | 88.7  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 26.3  | 26.6  | 26.0  | 23.1  | 20.0  | 18.7  | 16.3  | 14.9  | 17.1  | 21.5  | 27.5  | 27.1  |
|                                                                                  |             | MCDBR  | 29.8  | 29.6  | 28.2  | 24.7  | 22.2  | 20.5  | 18.6  | 17.1  | 19.6  | 23.4  | 28.0  | 29.4  |
|                                                                                  |             | MCWBR  | 11.9  | 11.0  | 10.7  | 11.2  | 9.7   | 6.7   | 5.5   | 5.6   | 6.3   | 8.6   | 9.4   | 10.9  |
|                                                                                  | 5% WB       | MCDBR  | 28.0  | 26.9  | 23.7  | 20.3  | 19.6  | 18.5  | 16.1  | 15.4  | 17.3  | 19.5  | 25.2  | 28.5  |
|                                                                                  |             | MCWBR  | 12.1  | 11.0  | 11.9  | 11.0  | 8.5   | 7.1   | 6.5   | 6.6   | 7.0   | 7.8   | 10.5  | 11.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.601 | 0.648 | 0.730 | 0.739 | 0.721 | 0.690 | 0.623 | 0.590 | 0.580 | 0.627 | 0.552 | 0.542 |
|                                                                                  | taud        |        | 1.692 | 1.563 | 1.429 | 1.500 | 1.535 | 1.606 | 1.787 | 1.850 | 1.860 | 1.714 | 1.765 | 1.785 |
|                                                                                  | Ebn at Noon |        | 224   | 220   | 207   | 205   | 205   | 209   | 225   | 235   | 238   | 222   | 235   | 235   |
|                                                                                  | Edh at Noon |        | 75    | 89    | 103   | 96    | 91    | 83    | 70    | 67    | 66    | 75    | 69    | 67    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 1781  | 1937  | 2038  | 2069  | 2023  | 1922  | 1842  | 1776  | 1917  | 1917  | 1830  | 1733  |
|                                                                                  | RadStd      |        | 70    | 81    | 102   | 84    | 58    | 76    | 51    | 69    | 64    | 56    | 42    | 54    |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           |       | +0.67   | +1.69  | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (2 neighbors) |       | +0.67   | +1.69  | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page